When we physicists explore this spacetime arena by experiments and by mathematics, we discover that space and time are unified in several ways. At the simplest level, when we look out in space, we are automatically looking backward in time because of how long it takes light to reach us. We see a quasar a billion light-years away as it was a billion years ago, when the light that enters our telescope was launched on its journey to us.

At a much deeper level, if you move relative to me at high speed, then we disagree on what events are simultaneous. You may think that two explosions, one on the Sun and the other on Earth are simultaneous, while I think the Earth explosion was five minutes before the one on the Sun. In this sense, what you regard as purely spatial (the separation of the explosions) I see as a mixture of space and time.

This mixing of space and time may seem counterintuitive, but it is fundamental to the very fabric of our universe. Fortunately, we can pretty much ignore it in this book except for Chapter 30.

The Bulk: Is It Real?



Throughout this book, I visualize warped space by picturing our universe as a two-dimensional warped membrane, or brane, that resides in a bulk with three space dimensions—as in Figure 21.1, for example. Of course, in reality our brane has three space dimensions and the bulk has four, but I’m not very good at drawing that, so in my pictures I usually throw one dimension away.